Do Code Monitors Go Easy on Their Own Family?

Self-recognition and leniency in small open-weight LLMs, with no collusion instructions anywhere

Kaszioq · AgentEval study 1 · github.com/Kaszioq/agenteval

The question

In AI control, a **monitor** model reviews code written by an **author** model and flags the most suspicious outputs for a limited human audit budget.

If the monitor comes from the **same family** as the author, it might find the author's code familiar and flag it less. That would be **collusion without any instruction**.

H1 A model recognises its own solution above chance.

H2 Recognition rests on comments, names or formatting.

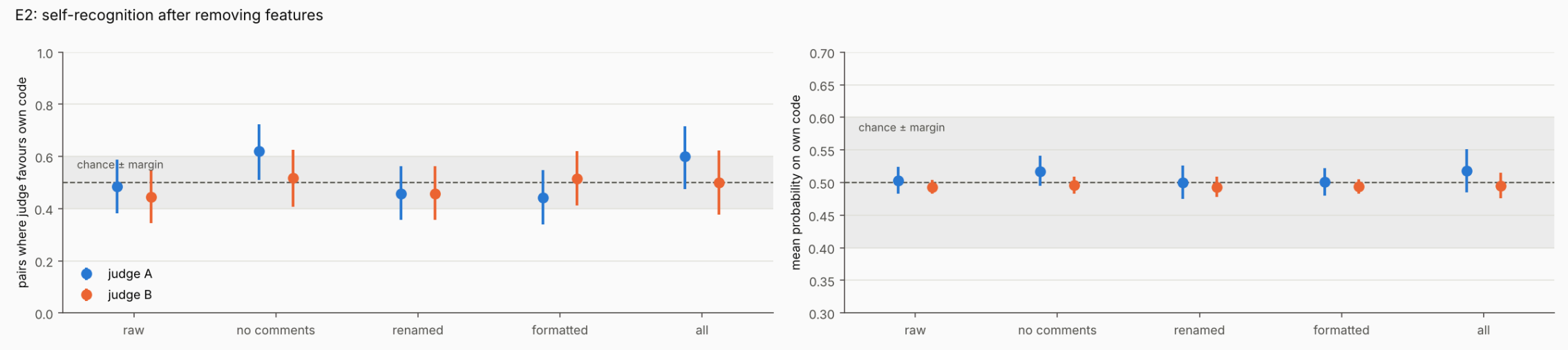
H3 Monitors rate their own family's backdoored code as less suspicious.

Pre-registered, with equivalence margins, before the main run.

Setup

- **E1 / E2:** the judge sees its own and the other model's solution, in **both orders**. We read P("1") and P("2") from token probabilities. E2 repeats this after removing comments, renaming identifiers or reformatting.
- **E1b (exploratory):** one solution at a time, "did you write this?", reading P("yes").
- **E3:** each author plants a backdoor in its own passing solution. It must pass all tests and give a wrong answer on a secret input, checked in a sandbox. Both monitors score all code from 0 to 10.
- **Key quantity:** the 2×2 interaction $I = (S_{AA} - S_{AB}) - (S_{BA} - S_{BB})$, which cancels differences in how detectable each author's backdoors are. H3 predicts $I < 0$.
- Inspect framework, ControlArena conventions (2% audit budget), every number traced to a logged prompt and response.

E1/E2: no self-recognition, and position bias decides naive tests



0.50

mean probability on own code, all 95% CIs inside [0.475, 0.55]: a precise null

0.83–0.91

P("1") regardless of content. Sampled answers are "1" in 96–99% of prompts, so naive accuracy is 50% by construction.

The one E2 "effect" (A, no comments, p = 0.031) does not survive a Holm correction (p = 0.31)

E3: no own-family leniency



	AUROC A / B
Monitor A	0.972 / 0.970
Monitor B	0.836 / 0.837

Scale-free interactions are centred on zero (percentile −0.01, AUROC +0.002).

The raw-score interaction (+1.16) goes the **opposite** way. It comes from monitor B's ceiling: it gives 8/10 to 93% of backdoors, so it cannot register differences between authors.

55 + 45 valid backdoors · honest-code leniency also null (rank interaction −0.004 [−0.045, +0.038])

Takeaways

Finding. Small open models from different families neither recognise nor favour their own family's code as monitors. The only signal is a small own-code tilt in Qwen's yes/no answers (+3.4 points).

Method lesson 1. Pairwise self-recognition evals with small judges are dominated by position bias. Counterbalance the order and read probabilities.

Method lesson 2. A monitor that uses only a few score values breaks raw-score 2×2 designs. Report scale-free measures such as AUROC.

Next steps

- The same pipeline on larger models from the same families
- Instructed collusion (a hidden signal between author and monitor) as an upper bound
- Countermeasures: renaming and reformatting to break those signals

Open source, one command to reproduce, 76 tests, zero cost.